SynthMicro — Whole Blood Smear XAI Report

Analysis ID: d4082c43a0c7457198d5b455beebd852
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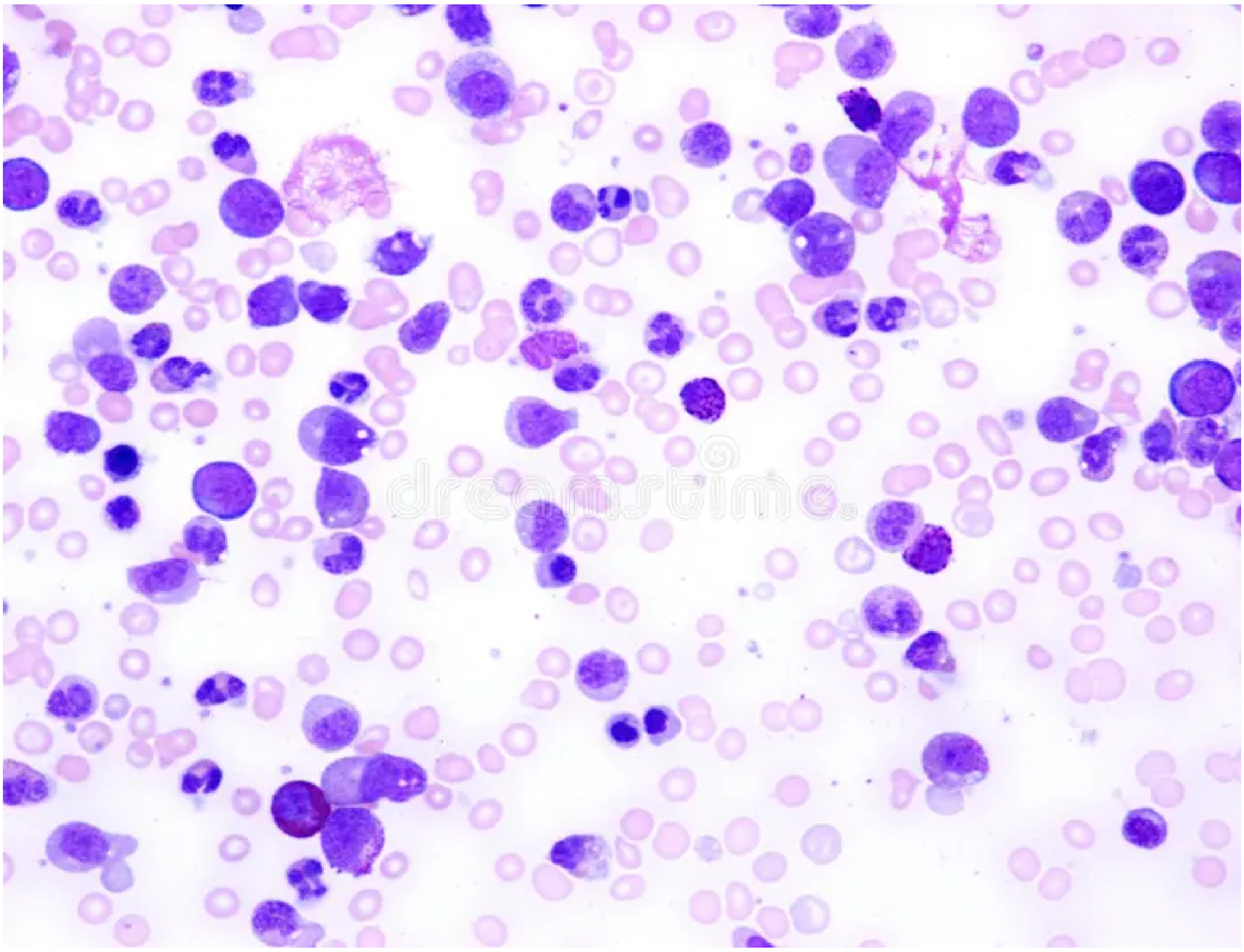
Pipeline: Cellpose segmentation → 384×384 cell extraction → ResNet50 classification → Grad-CAM → SHAP → explanatory report.

Measurement	Result
Cellpose objects	189
Nucleated-cell candidates	129
Myeloblast-like candidates	27
Myeloblast-like proportion	20.93%

Screening interpretation: the proportion above is a research-oriented model indicator based on candidate cells selected by the notebook's purple-score rule. It is not a cancer probability and does not establish AML or another diagnosis.

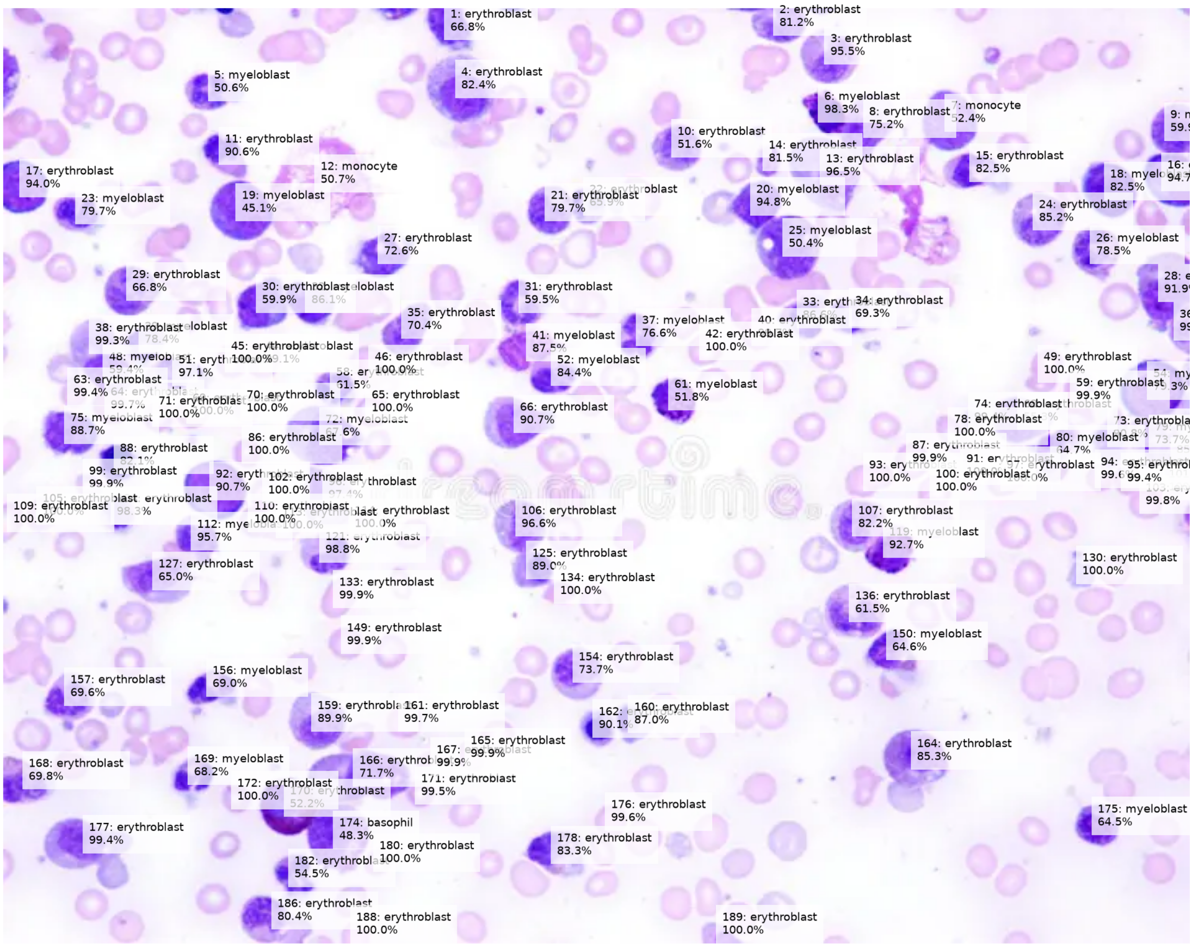
Whole Smear



Original uploaded blood-smear image analyzed by Cellpose.

Segmentation and Candidate Overlay

Whole Smear — Candidate Cell Classification



Cell-Level Results

ID	Area	Purple	Class	Confidence
1	643	110.4	erythroblast	66.8%
2	651	96.5	erythroblast	81.2%
3	1081	82.5	erythroblast	95.5%
4	1569	86.8	erythroblast	82.4%
5	676	92.3	myeloblast	50.6%
6	563	120.8	myeloblast	98.3%
7	1110	107.3	monocyte	52.4%
8	1088	102.7	erythroblast	75.2%
9	710	113.1	myeloblast	59.9%
10	758	94.1	erythroblast	51.6%
11	560	102.2	erythroblast	90.6%
12	2686	57.0	monocyte	50.7%
13	1647	86.3	erythroblast	96.5%
14	252	90.6	erythroblast	81.5%
15	743	87.2	erythroblast	82.5%
16	859	122.2	erythroblast	94.7%
17	865	120.2	erythroblast	94.0%
18	994	122.4	myeloblast	82.5%
19	1245	115.1	myeloblast	45.1%
20	728	118.2	myeloblast	94.8%
21	735	96.1	erythroblast	79.7%
22	440	97.0	erythroblast	65.9%
23	688	89.7	myeloblast	79.7%
24	972	87.5	erythroblast	85.2%
25	1404	99.9	myeloblast	50.4%
26	823	93.4	myeloblast	78.5%
27	875	91.3	erythroblast	72.6%
28	1357	102.0	erythroblast	91.9%
29	889	92.5	erythroblast	66.8%
30	895	106.8	erythroblast	59.9%
31	819	87.7	erythroblast	59.5%
32	694	113.3	myeloblast	86.1%
33	664	93.0	erythroblast	86.6%
34	715	86.1	erythroblast	69.3%
35	771	96.9	erythroblast	70.4%
36	257	90.1	erythroblast	99.8%
37	716	90.5	myeloblast	76.6%
38	683	58.6	erythroblast	99.3%
39	522	101.8	myeloblast	78.4%

ID	Area	Purple	Class	Confidence
40	246	38.7	erythroblast	99.7%
41	801	110.5	myeloblast	87.5%
42	317	35.5	erythroblast	100.0%
44	334	46.9	erythroblast	99.1%
45	290	40.1	erythroblast	100.0%
46	611	45.3	erythroblast	100.0%
48	626	110.5	myeloblast	59.4%
49	442	44.8	erythroblast	100.0%
51	813	80.4	erythroblast	97.1%
52	607	100.8	myeloblast	84.4%
54	1284	120.2	myeloblast	79.3%
58	468	97.2	erythroblast	61.5%
59	606	42.7	erythroblast	99.9%
61	674	125.9	myeloblast	51.8%
63	173	43.1	erythroblast	99.4%
64	374	54.7	erythroblast	99.7%
65	358	35.7	erythroblast	100.0%
66	1130	86.6	erythroblast	90.7%
68	895	94.3	erythroblast	55.3%
69	307	52.3	erythroblast	100.0%
70	189	41.7	erythroblast	100.0%
71	293	42.1	erythroblast	100.0%
72	1420	96.6	myeloblast	67.6%
73	662	95.6	erythroblast	90.8%
74	151	47.4	erythroblast	99.4%
75	826	107.5	myeloblast	88.7%
78	389	35.9	erythroblast	100.0%
79	780	98.7	myeloblast	73.7%
80	797	92.4	myeloblast	64.7%
85	464	122.3	erythroblast	98.4%
86	253	43.9	erythroblast	100.0%
87	416	39.4	erythroblast	99.9%
88	515	91.0	erythroblast	82.1%
91	295	37.9	erythroblast	100.0%
92	1206	113.8	erythroblast	90.7%
93	511	39.8	erythroblast	100.0%
94	230	38.8	erythroblast	99.6%
95	259	36.8	erythroblast	99.4%
96	1071	91.6	erythroblast	97.4%
97	362	37.6	erythroblast	100.0%
99	194	37.7	erythroblast	99.9%

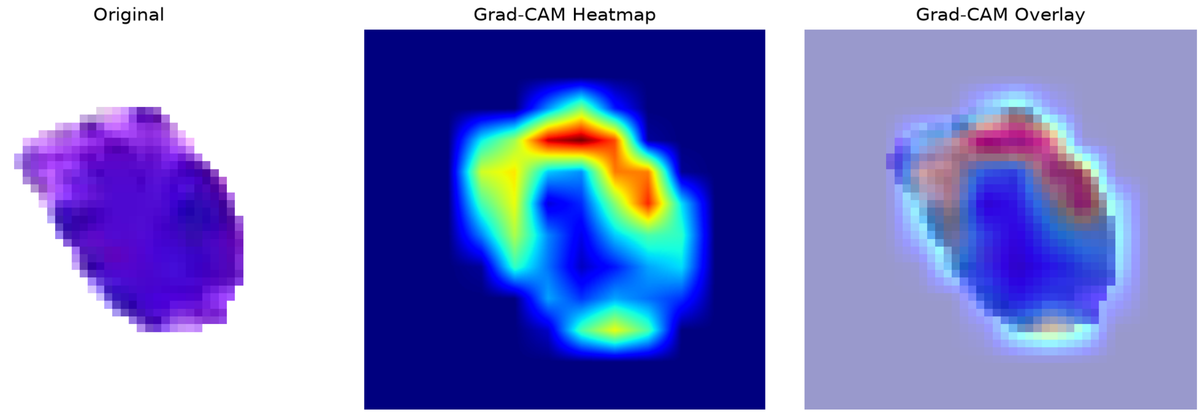
ID	Area	Purple	Class	Confidence
100	316	35.6	erythroblast	100.0%
101	191	49.3	erythroblast	100.0%
102	310	35.4	erythroblast	100.0%
103	330	41.1	erythroblast	99.8%
104	479	99.9	erythroblast	98.3%
105	317	36.2	erythroblast	100.0%
106	925	83.1	erythroblast	96.6%
107	977	85.2	erythroblast	82.2%
109	211	36.1	erythroblast	100.0%
110	289	36.3	erythroblast	100.0%
112	698	93.6	myeloblast	95.7%
113	277	41.2	erythroblast	100.0%
114	197	49.0	erythroblast	100.0%
119	752	127.1	myeloblast	92.7%
121	748	84.6	erythroblast	98.8%
125	518	90.5	erythroblast	89.0%
127	1093	86.7	erythroblast	65.0%
130	241	40.7	erythroblast	100.0%
133	451	39.8	erythroblast	99.9%
134	214	47.9	erythroblast	100.0%
136	1073	84.3	erythroblast	61.5%
149	351	39.0	erythroblast	99.9%
150	632	103.9	myeloblast	64.6%
154	916	83.4	erythroblast	73.7%
156	577	94.1	myeloblast	69.0%
157	776	90.4	erythroblast	69.6%
159	1075	78.6	erythroblast	89.9%
160	499	82.9	erythroblast	87.0%
161	392	41.7	erythroblast	99.7%
162	459	83.4	erythroblast	90.1%
164	1220	93.5	erythroblast	85.3%
165	238	37.5	erythroblast	99.9%
166	1805	99.8	erythroblast	71.7%
167	369	41.6	erythroblast	99.9%
168	745	89.1	erythroblast	69.8%
169	490	97.4	myeloblast	68.2%
170	1111	120.7	erythroblast	52.2%
171	214	46.4	erythroblast	99.5%
172	214	45.0	erythroblast	100.0%
174	1418	112.6	basophil	48.3%
175	587	108.1	myeloblast	64.5%

ID	Area	Purple	Class	Confidence
176	268	39.7	erythroblast	99.6%
177	1301	81.5	erythroblast	99.4%
178	903	71.9	erythroblast	83.3%
180	240	39.6	erythroblast	100.0%
182	564	94.0	erythroblast	54.5%
186	1009	81.3	erythroblast	80.4%
188	150	36.4	erythroblast	100.0%
189	195	76.3	erythroblast	100.0%

Grad-CAM Explanations

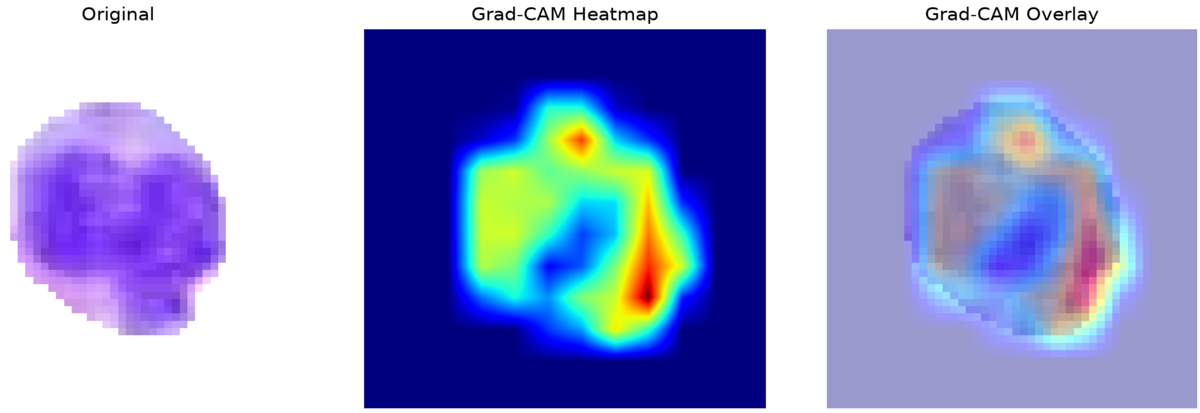
Grad-CAM shows regions contributing to the selected model class. It is an explanation of model behavior, not a direct measurement of a biological structure.

Cell 6 — myeloblast (98.3%)



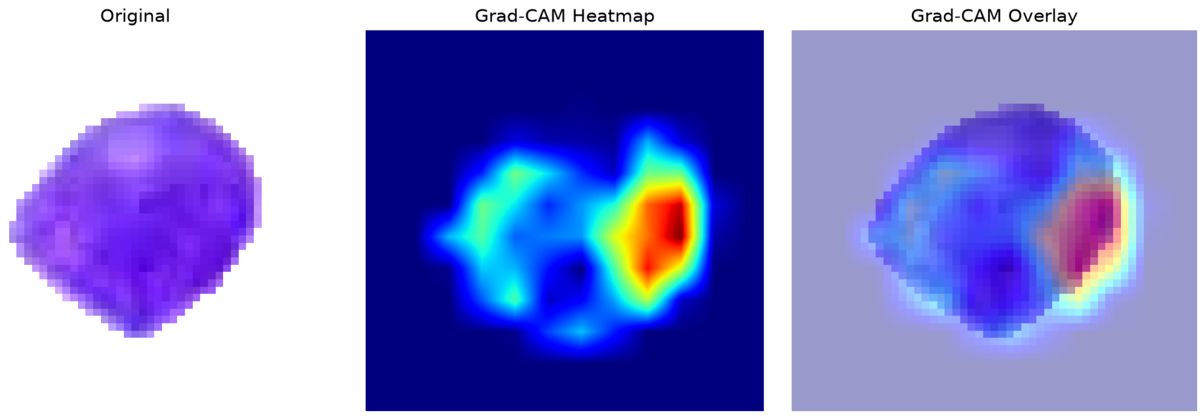
The ResNet50 model classified this segmented candidate as a myeloblast with 98.3% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 112 — myeloblast (95.7%)



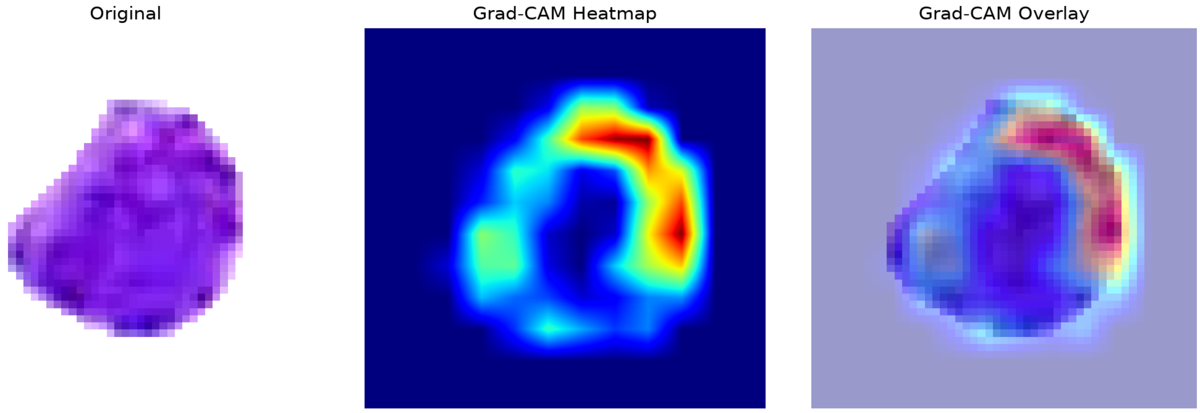
The ResNet50 model classified this segmented candidate as a myeloblast with 95.7% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 20 — myeloblast (94.8%)



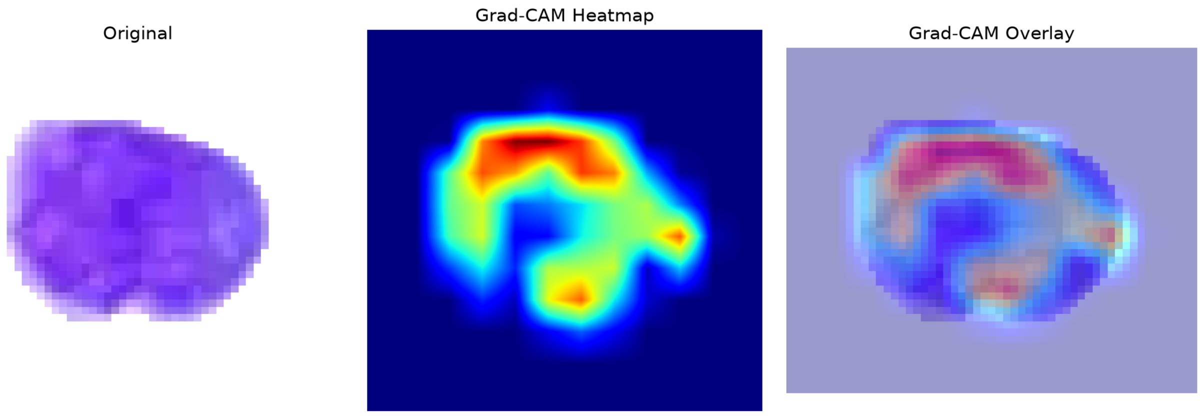
The ResNet50 model classified this segmented candidate as a myeloblast with 94.8% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 119 — myeloblast (92.7%)



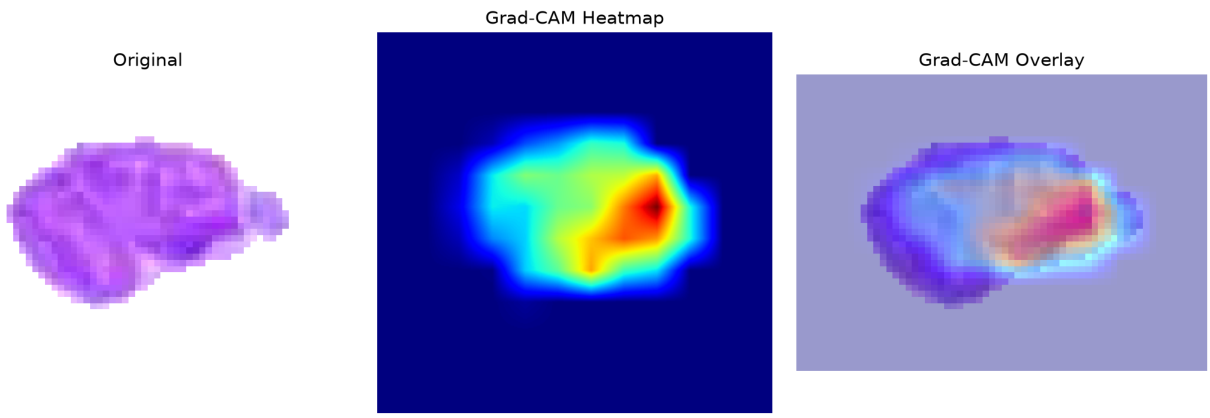
The ResNet50 model classified this segmented candidate as a myeloblast with 92.7% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 75 — myeloblast (88.7%)



The ResNet50 model classified this segmented candidate as a myeloblast with 88.7% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

Cell 41 — myeloblast (87.5%)



The ResNet50 model classified this segmented candidate as a myeloblast with 87.5% confidence. This is a model-level cell classification; a myeloblast-like prediction alone does not establish AML or another cancer diagnosis.

SHAP Explanations

SHAP attribution magnitude summarizes the magnitude of pixel-level contribution toward the selected class. It is a model explanation, not a clinical measurement.

SHAP output was unavailable for this analysis.

Interpretation and Limitations

• The classifier has five closed-set classes: basophil, erythroblast, monocyte, myeloblast and segmented neutrophil.

• RBCs and unknown cell types are not explicit classes and can therefore be forced into one of the five outputs.

• Cellpose segmentation quality affects every downstream result.

• Stain, microscope, illumination and acquisition differences can cause domain shift.

• Confidence is not a calibrated probability of disease.

• A myeloblast-like prediction is not equivalent to an AML diagnosis.

• Clinical assessment requires appropriate hematopathology and laboratory testing.